

KCH-MedRank: Knowledge-Constrained Learning-to-Rank for Evidence-Grounded Medical Retrieval-Augmented Generation

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Abstract

Biomedical evidence retrieval for medical question answering suffers from a fundamental limitation: standard pairwise query-passage scoring cannot capture the higher-order domain relationships that organize clinical knowledge. A passage about the thyroid hormone analog DITPA and a passage about levothyroxine share almost no surface lexical overlap, yet they belong to the same Medical Subject Headings (MeSH) subtree—a shared relationship invisible to any pairwise scoring function.

We introduce KCH-MedRank, a knowledge-constrained learning-to-rank framework for biomedical evidence reranking. The method augments a reproducible hybrid retrieval pipeline (BM25, dense retrieval, and Reciprocal Rank Fusion) with structured biomedical knowledge features derived from MeSH hierarchy annotations, biomedical entity linking, and the PrimeKG knowledge graph, organized through flat, pairwise graph, and local hypergraph architectural variants. Rather than treating structural signals as a standalone scoring formula, all features are fed into query-grouped LambdaMART—a tree-based learning-to-rank algorithm—yielding an interpretable, white-box reranker.

A key structural ablation reveals our central finding: **flat biomedical knowledge features** (MeSH hierarchy signals, entity overlap, and PrimeKG relations) already carry most of the retrieval gains without requiring any graph topology. Raw hypergraph structure applied *without* domain-grounded concept annotations introduces noise and degrades ranking. However, incorporating MeSH, entity, and PrimeKG constraints into the hypergraph reverses this noise into a statistically significant signal ($p < 0.01$). On the BioASQ held-out test split, KCH-MedRank achieves significant Recall@10 improvement over a true MedCPT Cross-Encoder baseline (+0.0157, $p = 0.0076$) while running $18.6\times$ faster at reranking time. Interpretability analysis traces all 648 rescued gold passages to three knowledge-grounded mechanisms: MeSH hierarchy paths (75.9%), shared entity clusters (23.9%), and PrimeKG relations (remainder). External validation on BEIR NFCorpus and PubMedQA provides supplementary robustness evidence. We position KCH-MedRank as a medical knowledge-grounded evidence selection module rather than as a clinical decision system.

1 Introduction

Ask a biomedical search engine “What is the function of DITPA?” and it retrieves dozens of passages about thyroid hormone analogs. A clinician scanning those results spots the connection immediately: DITPA is a thyroid hormone analog, and its mechanism sits under the Medical Subject Headings (MeSH) tree node for thyroid hormones. A standard retrieval pipeline scores each passage against the query by lexical overlap or embedding similarity, placing the most relevant evidence at rank 54. The passage is present in the candidate pool—but pairwise scoring alone cannot surface it.

Evidence-grounded medical question answering requires systems to retrieve relevant biomedical evidence before producing or selecting an answer. Retrieval-augmented generation (RAG) provides a general framework for combining retrieved evidence with language models [9], and recent medical benchmarking studies confirm that retrieval component selection critically shapes medical QA performance [17]. Yet the dominant paradigm—scoring each passage independently against the query through lexical or dense embeddings—suffers from a structural blind spot: biomedical evidence is frequently organized through higher-order relations among questions, passages, MeSH concepts, biomedical entities, documents, and knowledge-base entries. Pairwise query-passage comparisons cannot trace these multi-hop domain pathways.

The Problem

Across 943 held-out BioASQ test questions, a strong Hybrid RRF retriever leaves every gold evidence passage outside the top 10 for over 6% of queries where that evidence does exist somewhere in the top 100. This gap persists because biomedical evidence lives in higher-order relations. A passage about DITPA and a passage about levothyroxine share no surface lexical overlap and sit far apart in embedding space—yet both belong to the same MeSH subtree for thyroid hormones. A pairwise scoring function never asks about that shared membership.

Existing biomedical retrievers [6] and medical RAG systems [17, 18] score each passage independently against the query. They do not check whether two candidate passages share a MeSH ancestor, nor whether the biomedical entities extracted from a question form a dense cluster with those of a lower-ranked passage. Recent GraphRAG-style approaches [4, 10, 16] apply graph structure to retrieval, but their edges are typically derived from lexical co-occurrence or embedding similarity rather than from domain-grounded concept annotations such as MeSH, biomedical entity linking, or curated knowledge graphs like PrimeKG. Critically, no existing work isolates whether raw graph topology—stripped of medical knowledge—helps or actively *hurts* biomedical evidence ranking.

Our Approach

We introduce KCH-MedRank to answer precisely that question. The method builds on a reproducible candidate pool constructed with BM25 [12], dense retrieval, and Reciprocal Rank Fusion (RRF) [3]. It then enriches each query’s local candidate neighborhood with four categories of features: (1) retrieval scores from lexical, dense, and fused sources; (2) biomedical semantic features from MeSH hierarchy annotations and entity overlap metrics; (3) relation features from the PrimeKG biomedical knowledge graph; and (4) optional structural features from a local hypergraph that captures n-ary concept interactions that decompose into information loss under pairwise graph representations. All features are fed into query-grouped LambdaMART—a gradient-boosted tree-based learning-to-rank algorithm [1]—yielding a transparent, feature-attributable reranker.

We design three controlled architectural variants to isolate the contribution of structural topology from that of domain knowledge:

- (1) **Flat Knowledge LTR**: retrieval, MeSH, entity, and relation features with *no* graph structure whatsoever;
- (2) **Pairwise Graph LTR**: the same knowledge features plus pairwise graph edges connecting entities anchored to the query;
- (3) **KCH-MedRank (Local Hypergraph)**: knowledge features plus local hypergraph diffusion that preserves multi-entity interactions across hyperedges.

The core experimental design isolates three research questions:

- **RQ1:** Do flat biomedical knowledge features (MeSH hierarchy, entity overlap, PrimeKG relations) alone improve retrieval over pure retrieval signals, without any graph topology?
- **RQ2:** Does hypergraph structure help, or does raw topology introduce noise that degrades ranking?
- **RQ3:** Can medical knowledge constraints (MeSH annotations and entity grounding) reverse structural noise into a useful ranking signal?

Key Findings

A flat biomedical knowledge LTR baseline without graph structure already captures most of the aggregate gains over retrieval-only baselines, so our main performance claim is attributed to the combined knowledge-constrained learning-to-rank framework rather than to hypergraph topology alone. A controlled structural ablation reveals that raw hypergraph topology, applied without MeSH or entity annotations, degrades nDCG@10 by 0.0074 ($p = 0.0002$) relative to the flat knowledge model. However, adding MeSH, entity, and PrimeKG constraints to the same hypergraph reverses this degradation into a statistically significant improvement ($p < 0.01$). The hypergraph architecture provides a modest additional benefit over pairwise graph decomposition mainly at $k=5$; at $k=10$ the difference is not statistically significant. On the BioASQ held-out test split, KCH-MedRank achieves significant Recall@10 improvement over a true MedCPT Cross-Encoder baseline (+0.0157, $p = 0.0076$) while running $18.6\times$ faster at reranking time. An interpretability analysis traces all 648 rescued gold passages to three knowledge-grounded mechanisms: MeSH hierarchy paths account for 75.9% of rescues, shared entity clusters for 23.9%, and PrimeKG relations for the remainder.

Contributions

- A reproducible two-stage biomedical evidence retrieval pipeline with BM25, dense retrieval, Hybrid RRF, enhanced candidate fusion, and deterministic held-out evaluation.
- A knowledge-constrained learning-to-rank reranker that jointly exploits retrieval scores, biomedical semantic features, MeSH hierarchy signals, entity annotations, PrimeKG relations, and optional hypergraph structural features under query-grouped LambdaMART.
- A controlled structural ablation framework with three architectural variants (flat, pairwise graph, local hypergraph) that isolates the effect of domain knowledge from that of graph topology, demonstrating that medical knowledge constraints are *necessary* for structural signals to avoid becoming harmful noise.
- Interpretable post-hoc feature attribution showing that MeSH hierarchy paths drive 75.9% of rescued gold evidence passages, providing specific guidance for biomedical retrieval system design.
- Evidence-oriented evaluation across BioASQ, PubMedQA, and BEIR NFCorpus including Recall@k, Hit@k, MRR@k, nDCG@k, hard reranking subsets, paired bootstrap significance tests (10,000 resamples), failure analysis, and a $18.6\times$ speed advantage over Cross-Encoder reranking.

Scope and Limitations

We position KCH-MedRank as a focused evidence selection module rather than a complete clinical QA system. The method does not replace clinical judgment, provide clinical diagnosis, or guarantee answer correctness. The failure analysis reveals cases where gold passages remain lost, and the hypergraph benefit over flat knowledge features is modest and concentrated at early ranks ($k \leq 5$). In the discussion, we characterize the method’s remaining failure modes and identify directions for future work, including broader knowledge graph coverage and integration with generation-level evaluation.

2 Related Work

2.1 Medical Retrieval-Augmented Question Answering

Medical RAG benchmarking has recently made systematic evaluation of corpora, retrievers, and backbone models more visible. The MedRAG/MIRAGE benchmark evaluates medical RAG configurations over multiple QA datasets and highlights that retrieval settings have a large effect on downstream medical QA [17]. Iterative medical RAG extends this direction by generating follow-up retrieval queries for complex questions [18]. This work is narrower: it focuses on evidence retrieval and reranking before making strong generation claims.

2.2 Biomedical QA Datasets and Biomedical Retrieval

BioASQ provides a long-running biomedical semantic indexing and QA benchmark [14], and BioASQ-QA offers manually curated questions, reference answers, documents, snippets, and concepts for biomedical QA research [8]. PubMedQA is a complementary biomedical research question answering dataset with yes/no/maybe labels over PubMed abstracts [5]. BEIR provides a heterogeneous information retrieval benchmark that includes biomedical subsets such as NFCorpus [13]; here it is used only for an external retrieval-only diagnostic. For biomedical semantic retrieval, MedCPT uses large-scale PubMed search logs to train a biomedical retriever and reranker [6].

2.3 Graph and Hypergraph RAG

GraphRAG-style methods organize retrieved information using graph structure, with local-to-global search proposed for query-focused summarization [4]. Medical Graph RAG connects user documents, credible medical sources, and controlled vocabularies for medical response grounding [16]. HyperGraphRAG argues that hyperedges can represent n-ary relations beyond ordinary pairwise graph edges [10], and cross-granularity HGRAG models entities and passages at different retrieval granularities [15]. A recent Medical HyperRAG preprint studies patient-centered clinical QA with a ClinBridge hypergraph over records, cases, and guidelines [19]. KCH-MedRank follows the high-order relation motivation, but differs in scope: it uses public biomedical QA resources, avoids private clinical records, and evaluates a local evidence-reranking module rather than a full patient-centered clinical decision-support framework.

2.4 Controlled Biomedical Knowledge

MeSH is a controlled, hierarchically organized vocabulary maintained by the National Library of Medicine for indexing and searching biomedical information [11]. PrimeKG integrates multimodal precision-medicine relations across diseases, proteins, drugs, phenotypes, and other biomedical

Figure 1. Overview of KCH-MedRank

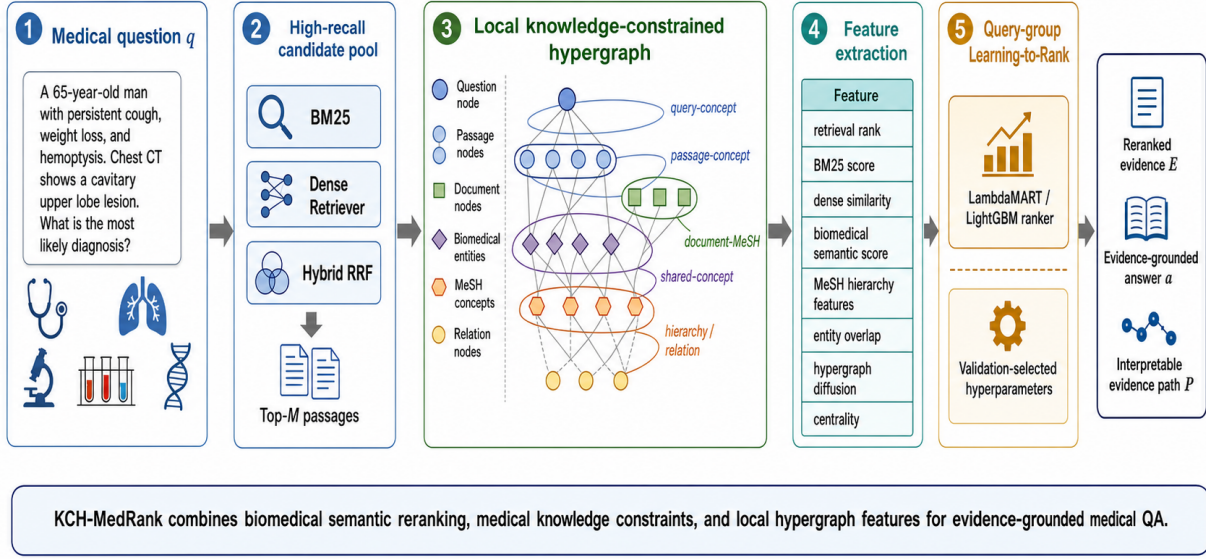


Figure 1: Overview of KCH-MedRank. The method first builds a high-recall candidate pool, constructs a local knowledge-constrained hypergraph around each question and candidate set, and uses hypergraph and biomedical semantic features in query-group learning-to-rank.

entities [2]. In this study, MeSH hierarchy features are central to the medical constraint design, while PrimeKG is treated as an auxiliary relation source because exact-name coverage is sparse.

3 Task Definition

Given a medical question q , a biomedical corpus C , and optional knowledge sources K , the system returns a ranked evidence list E and optionally an answer a . The primary task is evidence retrieval and reranking; answer generation is secondary.

For each question, a first-stage retriever returns a candidate pool $D_q = \{d_1, \dots, d_M\}$ from C . A reranker then estimates an improved ordering over D_q using retrieval scores, biomedical semantic scores, concept and entity features, and local hypergraph features. Evaluation emphasizes Recall@ k , MRR@ k , nDCG@ k , and evidence coverage.

The formal output is:

$$(a, E, P),$$

where a is optional in the evidence-focused setting, E is the ranked supporting evidence list, and P is an optional interpretable entity, concept, or hypergraph path used for case analysis.

4 Method

4.1 First-Stage Retrieval

The first stage retrieves a high-recall candidate pool before any knowledge-aware reranking is applied. We use BM25 as the lexical baseline, dense retrieval as the semantic baseline, and Reciprocal

Rank Fusion (RRF) as the default hybrid retriever. For a retrieval source s , let $r_s(q, d)$ be the rank of passage d for question q . The fused score is:

$$\text{RRF}(q, d) = \sum_{s \in S} \frac{w_s}{k + r_s(q, d)},$$

where S is the set of retrieval sources, w_s is a source weight, and $k = 60$ in the reported experiments. The original Hybrid RRF uses BM25 and dense retrieval with equal weights.

The enhanced candidate-generation path is designed to raise the top-100 ceiling for reranking. It adds fielded BM25 with title, text, and MeSH expansion fields, plus MedCPT dual-encoder dense retrieval as an additional biomedical semantic source. Fusion weights are selected on the validation split only. The selected w122 configuration gives BM25 and dense retrieval weight 1.0, fielded BM25 weight 1.2, and MedCPT dual-encoder retrieval weight 0.2. This setting is selected for validation Recall@100 rather than test performance, because reranking can improve only passages already present in the candidate pool.

4.2 Knowledge-Constrained Local Structure

For each question and its top- M candidates, KCH-MedRank builds feature variants with increasing structural assumptions. The flat knowledge variant uses retrieval, biomedical semantic, MeSH, entity, and relation features without any graph construction. The pairwise graph variant adds anchor-based edges among the same local objects. The hypergraph variant builds a local hypergraph $H_q = (V_q, \mathcal{E}_q)$ whose node set includes a question node, candidate passage nodes, document nodes, biomedical entity nodes, MeSH concept nodes, optional MeSH tree nodes, and optional relation nodes. The hyperedge set connects multiple nodes at once and therefore represents higher-order relations that are not naturally captured by pairwise similarity alone.

The main hyperedge types are question-entity, question-MeSH, question-passage candidate, passage-entity, passage-document, document-MeSH, shared-entity passages, entity-type concept, MeSH hierarchy, and auxiliary PrimeKG relation edges. Pairwise graph and no-medical-knowledge variants are constructed from the same candidate pool for ablation: the pairwise variant decomposes multi-node hyperedges into anchor-based pairwise edges, while the no-knowledge variant keeps only rank-band and candidate structure.

The local design avoids constructing a global graph over millions of biomedical passages. It also keeps the method interpretable: feature inspection and case studies can report which concept overlaps, hierarchy links, shared candidate clusters, or graph and hypergraph centrality signals influenced reranking.

4.3 Hypergraph Diffusion and Structural Features

KCH-MedRank uses hypergraph diffusion as an interpretable feature rather than as a standalone ranking formula. The seed distribution places mass on the question node, question entity nodes, and question MeSH nodes. At each iteration, scores are propagated through weighted hyperedges:

$$x^{(t+1)} = (1 - \alpha)x^{(0)} + \alpha \cdot \text{Propagate}(x^{(t)}, \mathcal{E}_q),$$

where $\alpha = 0.85$ and the implementation uses three diffusion iterations. Passage-level diffusion scores are then min-max normalized within each question candidate set. We also compute weighted degree centrality, local node and hyperedge counts, shared-entity edge counts, document-MeSH edge counts, and optional hierarchy-edge counts.

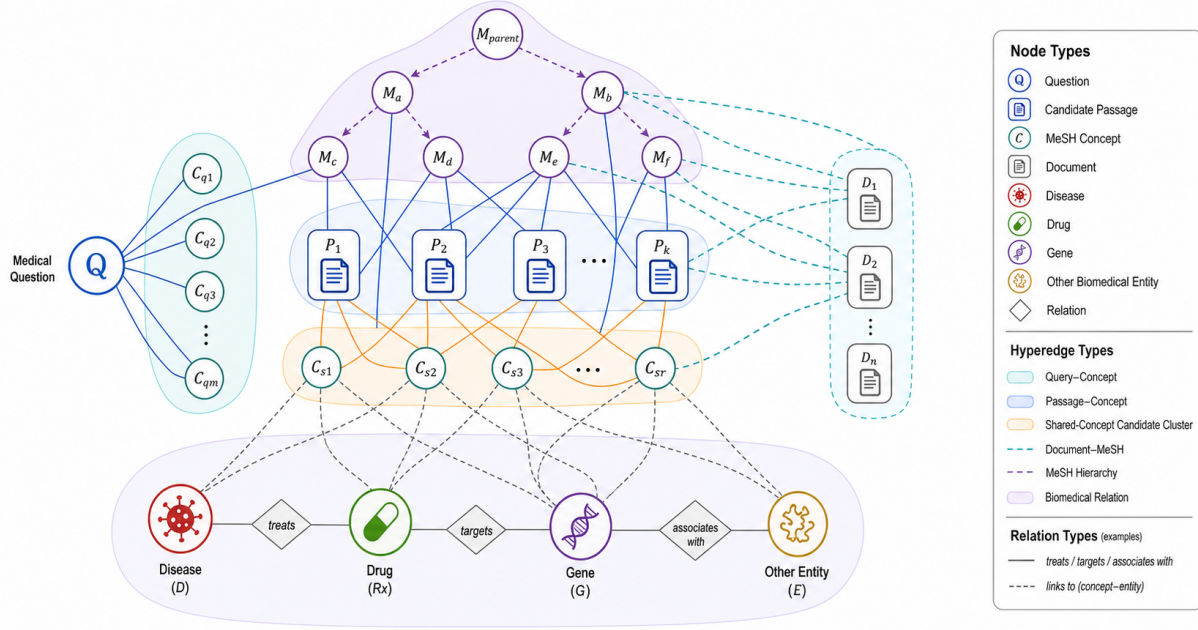


Figure 2: Local knowledge-constrained hypergraph structure around one question and its candidate passages. Nodes represent questions, passages, documents, biomedical entities, MeSH concepts, and auxiliary relation endpoints; translucent regions and bundled links indicate multi-node hyperedges used for interpretable ranking features.

4.4 Biomedical Knowledge Features

The reranker uses several medically motivated feature groups. Entity features include entity overlap count, entity Jaccard similarity, question entity coverage, and passage entity count. Exact MeSH features include MeSH overlap, MeSH Jaccard similarity, question MeSH coverage, and passage MeSH count. Hierarchy-aware MeSH features include exact descriptor overlap, parent match, ancestor match, sibling match, tree-distance similarity, query concept coverage, passage hierarchy coverage, passage concept specificity, and shared candidate clusters over MeSH terms or parent concepts.

PrimeKG-derived features are included only as auxiliary relation signals: relation count, question relation coverage, and local relation-edge count. Because exact-name relation coverage is sparse, these features are not treated as the main source of improvement.

Figure 3. KCH-MedRank Feature Extraction and Learning-to-Rank Reranking Module

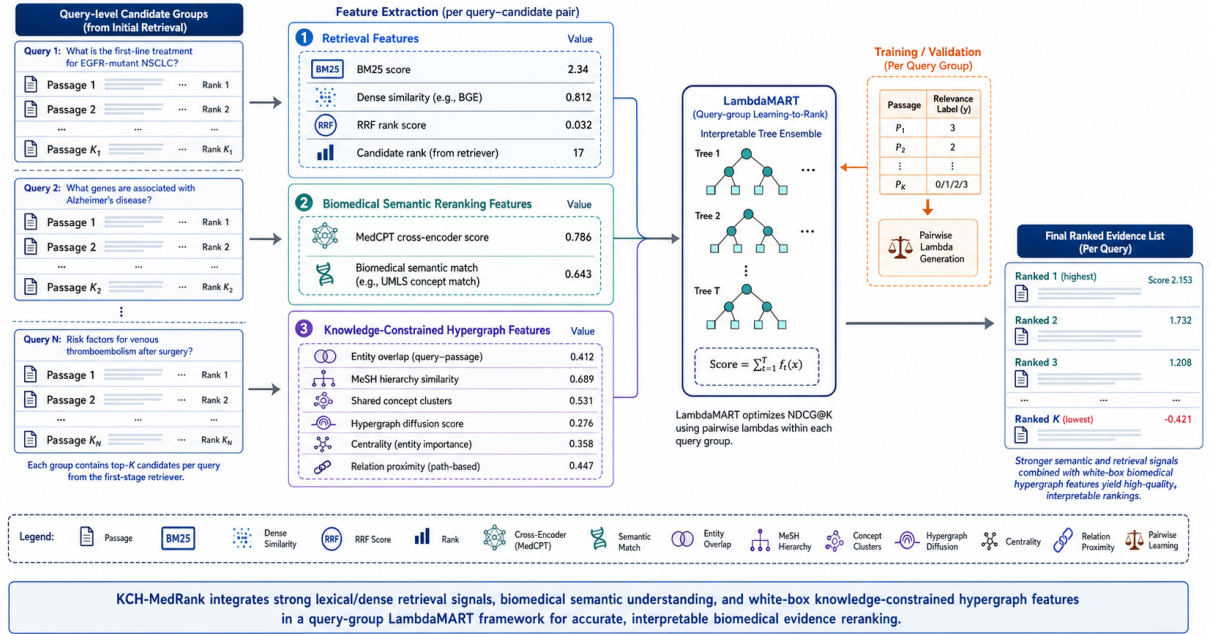


Figure 3: Feature extraction and query-group learning-to-rank module. Retrieval scores, cached biomedical semantic features, flat biomedical knowledge features, and optional knowledge-constrained graph or hypergraph features are combined in a LambdaMART-style ranker to produce the final evidence ordering. In the reported BioASQ KCH-MedRank runs, the semantic block is implemented from cached dense or dual-encoder predictions rather than online Cross-Encoder scoring.

Algorithm 1: KCH-MedRank local evidence control and reranking

- Candidate retrieval.** For question q , retrieve top- M passages from BM25, dense retrieval, fielded BM25, and MedCPT dual-encoder sources; fuse them with validation-selected weighted RRF.
- Local structure construction.** For the graph and hypergraph variants, create nodes for q , candidate passages, documents, extracted biomedical entities, MeSH descriptors, MeSH hierarchy concepts, and optional relation endpoints. Add edges or hyperedges for query-concept, passage-concept, document-MeSH, shared-concept candidate clusters, hierarchy links, candidate membership, and auxiliary PrimeKG relations. The flat knowledge baseline skips this step and uses only non-structural biomedical features.
- Hyperedge weighting and normalization.** Assign edge weights by source reliability and local evidence strength: candidate membership edges use normalized RRF scores; entity and MeSH edges use normalized overlap/coverage; hierarchy edges use inverse tree distance; relation edges are down-weighted because of sparse exact-name coverage. Within each local graph, incident edge weights are normalized before propagation.
- Diffusion propagation.** Initialize mass on the question, query-entity, and query-MeSH nodes. Run three restart diffusion iterations with $\alpha = 0.85$, then normalize passage diffusion scores within the candidate set.
- Passage feature extraction.** For each passage, concatenate retrieval scores and ranks, precomputed biomedical semantic retrieval features, entity overlap, MeSH exact and hierarchy features, auxiliary relation features, diffusion scores, centrality, and local graph-count features.

What drives the learned ranker?

Retrieval and semantic signals dominate; biomedical structure remains visible.

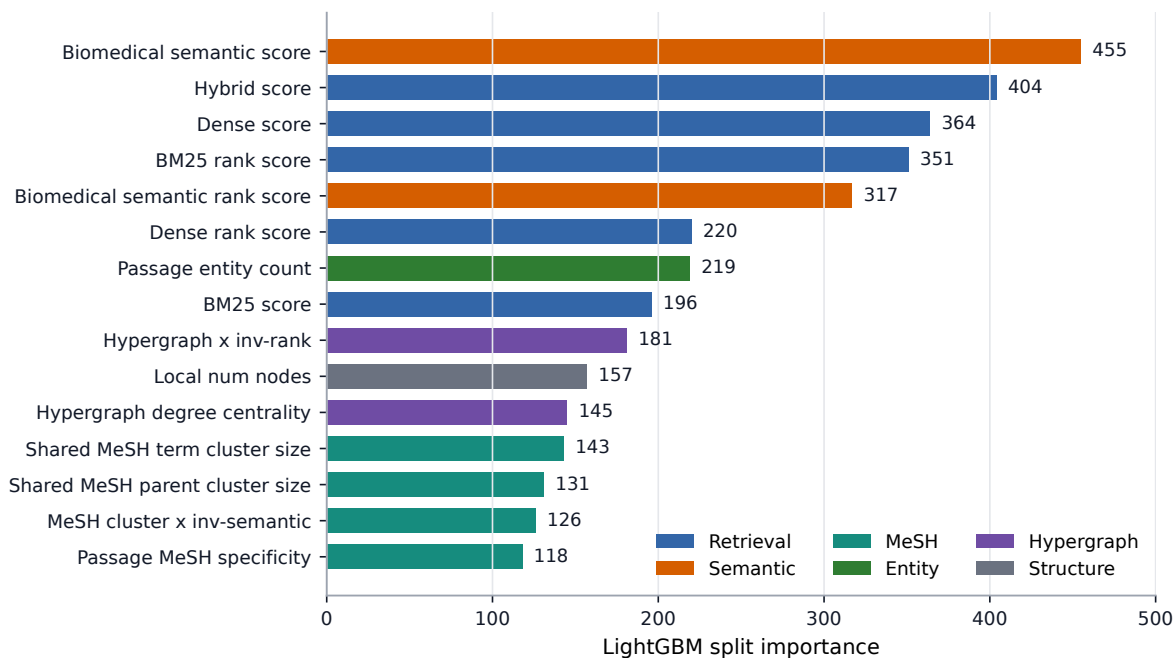


Figure 4: Top KCH-MedRank feature importances from the enhanced BioASQ experiment. Colors group features into retrieval, semantic, entity, MeSH, structural, and hypergraph families.

4.5 Learning-to-Rank

The main supervised reranker is a query-group learning-to-rank model: a LightGBM gradient boosted tree ranker trained with a LambdaMART-style objective [7]. Each question forms one ranking group, candidate passages form rows, and binary relevance labels are derived from gold evidence annotations. The feature vector combines retrieval scores, precomputed biomedical semantic retrieval features, entity features, exact and hierarchy-aware MeSH features, auxiliary relation features, and optional local graph or hypergraph structural features.

Hyperparameters are selected on the validation split. The reported search space is number of leaves $\{15, 31\}$, learning rate $\{0.03, 0.05\}$, estimators $\{80, 160\}$, and blend weight $\{0, 0.1, 0.2, 0.35\}$, where the blend weight interpolates model scores with the original candidate rank score. Selection prioritizes validation MRR@10, then Recall@10 and nDCG@10. After validation selection, the final model is trained on train plus validation questions and evaluated once on the held-out test split.

Feature importance in Figure 4 shows that semantic and first-stage retrieval signals remain central, while MeSH cluster, passage specificity, and hypergraph centrality features contribute interpretable biomedical and structural evidence. This supports a cautious claim: KCH-MedRank improves reranking by combining retrieval strength, biomedical semantics, flat biomedical knowledge features, and local structure signals, not by relying on one isolated knowledge source.

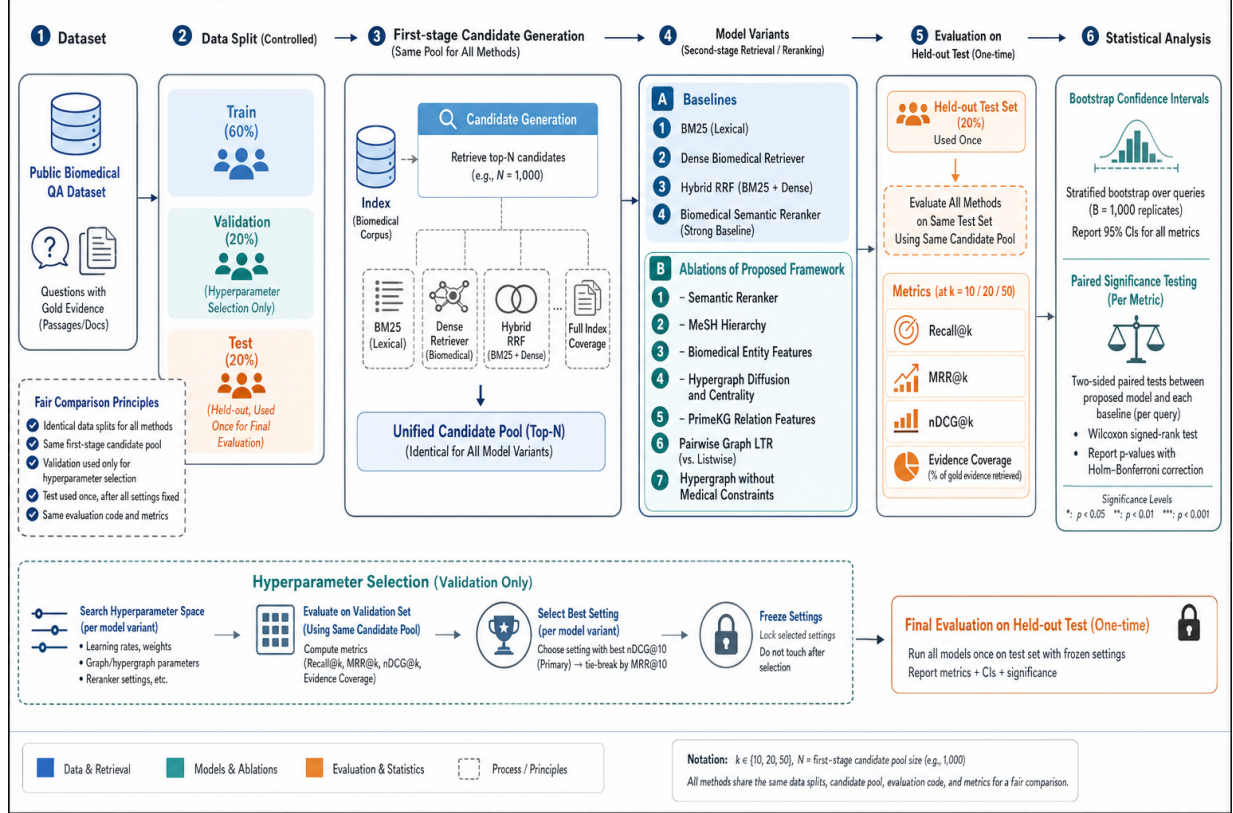


Figure 5: Controlled experimental design for candidate generation, reranking baselines, ablations, validation-only model selection, held-out test evaluation, and paired statistical testing. The diagram is schematic; the reported BioASQ experiments use validation-selected top-100 candidate pools and evaluate $k \in \{1, 3, 5, 10, 20, 50, 100\}$ as specified in the protocol.

5 Experimental Setup

5.1 Dataset

The primary dataset is `rag-datasets/rag-mini-bioasq`, used as a compact BioASQ-style biomedical evidence retrieval setting. The full processed set contains 4,719 questions. We use a deterministic question-level split based on question identifier buckets: training questions are all buckets except validation and test, validation uses bucket remainder 3, and test uses bucket remainder 4 under modulo 5. In the enhanced BioASQ experiment, this gives 2,832 training questions, 944 validation questions, and 943 held-out test questions. The split is fixed before model selection.

Each question is paired with a set of gold evidence passage identifiers. Evaluation is performed at the passage level. The reranker receives only candidates returned by the first-stage retrieval pool, so Recall@100 of the candidate generator is reported as an upper-bound analysis. The split is question-level, and passages from test questions are not used as positive training labels for the reranker. PubMedQA is used as a secondary diagnostic dataset rather than the main result source. BEIR NFCorpus is added as an external retrieval-only robustness diagnostic using its official train, validation, and test qrels; because the project-specific MeSH, entity, PrimeKG, and local graph/hypergraph features are not directly constructed for NFCorpus, this diagnostic compares BM25, dense retrieval, Hybrid RRF, and retrieval-feature LambdaMART only.

5.2 Baselines

We compare against both first-stage retrieval baselines and reranking baselines. The first-stage baselines are BM25, dense biomedical retrieval, original Hybrid BM25 + dense RRF, fielded BM25 with MeSH expansion, MedCPT dual-encoder retrieval, and enhanced Hybrid RRF. Enhanced Hybrid RRF is not tuned on the test split; its selected w122 weights are chosen by validation Recall@100.

The reranking baselines are MedCPT Cross-Encoder on the same enhanced candidate pool, retrieval-feature-only LambdaMART, flat biomedical knowledge LambdaMART without graph structure, pairwise graph learning-to-rank, hypergraph learning-to-rank without medical knowledge, and full KCH-MedRank. The flat knowledge baseline uses the same retrieval, biomedical semantic, entity, exact MeSH, hierarchy-aware MeSH, and PrimeKG relation features as the full model, but removes graph and hypergraph diffusion, centrality, local edge-count, and graph-interaction features. This baseline directly tests whether medical knowledge features alone explain the gain. Leave-one-feature-group ablations (removing the biomedical semantic score, MeSH hierarchy features, biomedical entity features, hypergraph diffusion and centrality, and PrimeKG relation features) are retained in the appendix as complementary diagnostics but are not the primary ablation evidence, because high multicollinearity among knowledge features leads to automatic compensation in tree-based models.

5.3 Evaluation

The main retrieval metrics are Recall@ k , Hit@ k , MRR@ k , nDCG@ k , and evidence coverage for $k \in \{1, 3, 5, 10, 20, 50, 100\}$. With multiple gold passages per question, Recall@ k is macro evidence coverage: for each question, it is the fraction of that question’s gold passages appearing in the top k , then averaged over questions. Hit@ k is query-level success: it is 1 if at least one gold passage appears in the top k and 0 otherwise. MRR@ k is also query-level: it is the reciprocal rank of the first gold passage within the top k , or 0 if no gold passage appears. nDCG@ k uses qrel relevance values as gains; because the current evidence labels are binary, it reduces to binary-gain DCG normalized by the ideal ordering of all available gold passages up to k . Evidence coverage is reported as the same macro passage recall value as Recall@ k to connect the metric to the medical evidence-grounding interpretation.

Statistical testing uses paired bootstrap over held-out test questions with 10,000 resamples and a fixed random seed. We report confidence intervals and two-sided p-values against original or enhanced Hybrid RRF and against the true MedCPT Cross-Encoder baseline when available. A hard reranking subset is used for diagnosis: it contains test questions for which Hybrid top-100 includes at least one gold passage but Hybrid top-10 misses all gold passages. This subset is not a replacement for the full held-out test evaluation.

5.4 Validation Selection

Table 1 reports the enhanced candidate-generation validation selection. The selected w122 setting uses BM25 and dense retrieval as base sources, fielded BM25 as a stronger recall source, and MedCPT dual-encoder retrieval as an additional biomedical semantic source. It is selected because it maximizes validation Recall@100, matching the reranking objective of improving the candidate ceiling.

The complete candidate-fusion search space contains the original Hybrid setting (1.0, 1.0, 0.0, 0.0) for BM25, dense, fielded BM25, and MedCPT dual encoder weights, plus w052 (1.0, 1.0, 0.5, 0.2), w084 (1.0, 1.0, 0.8, 0.4), w102 (1.0, 1.0, 1.0, 0.2), and w122 (1.0, 1.0, 1.2, 0.2). Only validation

Table 1: Validation-selected enhanced candidate generation. Fusion weights are selected only on the validation split.

Fusion run	BM25	Dense	Fielded BM25	MedCPT DE	Val. R@100	Test R@100
Original Hybrid	1.0	1.0	0.0	0.0	0.6325	0.6077
w052	1.0	1.0	0.5	0.2	0.6391	0.6163
w084	1.0	1.0	0.8	0.4	0.7329	0.7084
w102	1.0	1.0	1.0	0.2	0.7529	0.7285
w122, selected	1.0	1.0	1.2	0.2	0.7568	0.7388

Recall@100 is used to choose among these settings; the held-out test split is evaluated after this selection. The LambdaMART hyperparameter search space is specified in Section 4, and model selection is likewise validation-only.

5.5 Downstream Generation Pilot

To connect evidence selection with the medical RAG setting without expanding generation into the main claim, we retain a controlled PubMedQA pilot with DashScope’s OpenAI-compatible Qwen3-8B endpoint as an appendix-style diagnostic. The evaluation uses 100 PubMedQA questions from the common subset covered by Hybrid RRF, the available semantic cross-encoder PubMedQA output, and the KCH-style hypergraph reranker output. Each condition uses the same top-5 evidence passages, prompt template, temperature 0, maximum 256 output tokens, and no fine-tuning. The model must return one of **yes**, **no**, or **maybe** with passage citations. We report Accuracy, Macro-F1, citation support rate, unsupported claim rate, and answer-evidence entity consistency. This pilot is not a replacement for the BioASQ or NFCorpus retrieval evaluations.

5.6 Hyperparameter Sensitivity

KCH-MedRank introduces two hypergraph diffusion hyperparameters: damping d and iterations t . To assess sensitivity, we perform a correlation-based stability analysis on 30 randomly sampled held-out questions. For each question, we build the full knowledge hypergraph once and run diffusion with all (d, t) combinations from $d \in \{0.7, 0.8, 0.85, 0.9, 0.95\}$ and $t \in \{3, 5, 7, 10\}$. We compute the Pearson correlation between per-passage diffusion scores under each parameter pair and the baseline ($d = 0.85, t = 5$).

Figure 6 presents the results. For $t \geq 5$, per-passage diffusion scores are highly correlated with the baseline, with $r > 0.99$ for same-damping comparisons and $r \geq 0.89$ in all cases. At $t = 3$, diffusion is under-converged ($r \approx 0.51$) and produces substantially different passage scores. At $t = 10$, excessive iterations cause score compression ($r = 0.75\text{--}0.84$). The chosen defaults ($d = 0.85, t = 5$) fall within the stable region where diffusion scores show high linear correlation with the baseline. This correlation-level analysis suggests that hypergraph diffusion is robust to the exact (d, t) choice for $t \geq 5$, though full retrieval-metric stability under the complete LambdaMART pipeline remains to be systematically confirmed.

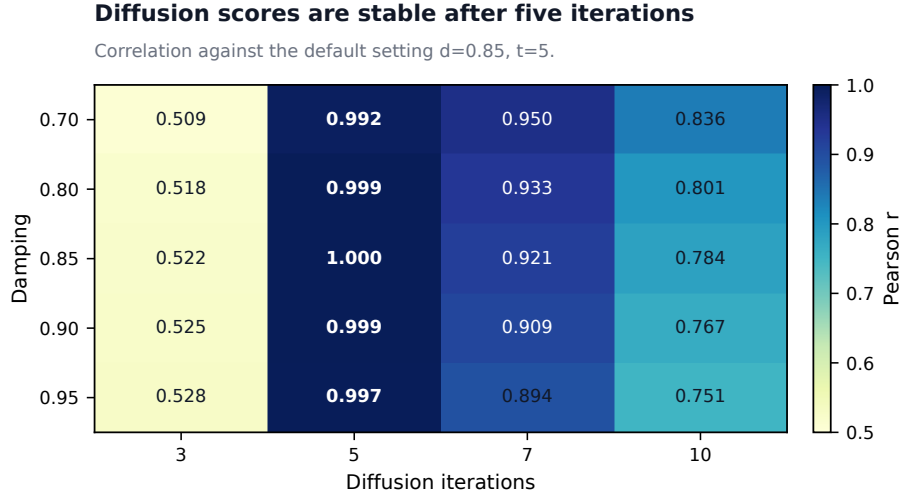


Figure 6: Hypergraph diffusion sensitivity. Each cell reports the Pearson correlation between per-passage diffusion scores under a damping–iteration pair and the default setting ($d = 0.85, t = 5$) on 30 sampled held-out questions.

Table 2: Held-out BioASQ evidence retrieval results for the enhanced candidate pool.

Method	Recall@10	MRR@10	nDCG@10	Recall@100
Original Hybrid RRF	0.4636	0.7550	0.5848	0.6077
Enhanced Hybrid w122	0.4660	0.7530	0.5844	0.7388
MedCPT Cross-Encoder	0.5172	0.7775	0.6390	0.7388
Enhanced KCH-MedRank	0.5329	0.7867	0.6433	0.7388

6 Results

6.1 Candidate Pool Quality

Table 2 and Figure 7 summarize the main held-out BioASQ retrieval results. The original Hybrid RRF test baseline has Recall@10 of 0.4636, MRR@10 of 0.7550, nDCG@10 of 0.5848, and Recall@100 of 0.6077. The enhanced w122 candidate pool was selected on validation Recall@100 and reaches test Recall@100 of 0.7388, providing a stronger candidate ceiling for reranking. This step is important because reranking cannot recover gold evidence that is absent from the candidate pool.

6.2 Main Held-Out Test Results

Enhanced KCH-MedRank reaches Recall@10 of 0.5329, MRR@10 of 0.7867, nDCG@10 of 0.6433, and Recall@100 of 0.7388. Relative to original Hybrid RRF, this corresponds to gains of +0.0693 Recall@10, +0.0317 MRR@10, and +0.0585 nDCG@10. Relative to the enhanced Hybrid RRF candidate ordering, KCH-MedRank improves Recall@10 by +0.0669, MRR@10 by +0.0337, nDCG@10 by +0.0589, and evidence coverage@10 by +0.0669.

Figure 8 reports paired bootstrap testing against enhanced Hybrid RRF. All four top-10 metrics are significant with $p = 0.0002$. This confirms that the improvement is not only a consequence of the stronger candidate pool: the reranking step itself changes the top-10 evidence ordering in a

BioASQ held-out evidence retrieval: KCH-MedRank improves top-10 ranking

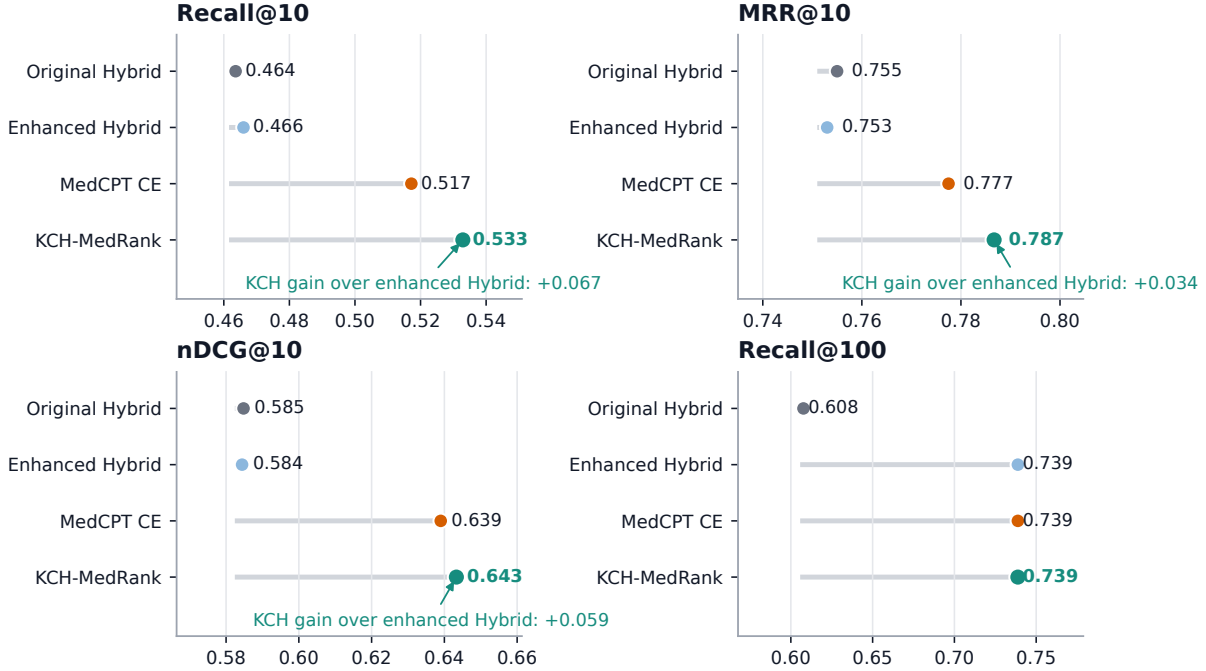


Figure 7: BioASQ held-out retrieval performance. The grouped bars make the candidate-pool ceiling and final top-10 reranking gains visible across Recall@10, MRR@10, nDCG@10, and Recall@100.

measurable way.

6.3 Comparison with Biomedical Semantic Reranking

On the same enhanced candidate pool, the true MedCPT Cross-Encoder baseline reaches Recall@10 of 0.5172, MRR@10 of 0.7775, and nDCG@10 of 0.6390. Figure 8 also reports paired bootstrap results against this strong semantic baseline. KCH-MedRank significantly improves Recall@10 (+0.0157, $p = 0.0076$), while MRR@10 and nDCG@10 are positive trends but not statistically significant.

The comparison with MedCPT Cross-Encoder is the stricter test for the proposed reranker. It shows that KCH-MedRank improves evidence coverage at top 10 beyond a strong biomedical semantic reranker. The rank-sensitive MRR and nDCG gains are positive trends but not significant, consistent with the interpretation that the method is particularly effective at recovering additional relevant evidence into the top-10 set. These results support the paper’s evidence-retrieval claim. They do not yet justify making answer generation the primary contribution.

6.4 Hard Reranking and PubMedQA Diagnostics

Figure 9 reports the diagnostic hard subset in which the enhanced Hybrid candidate pool contains gold evidence in the top 100 but misses all gold evidence in the top 10. This subset contains 65 held-out BioASQ questions and is not used as a replacement for the full test set. On this subset, enhanced Hybrid RRF has Recall@10 of 0 by construction, while full KCH-MedRank reaches Recall@10 of 0.4704 and the retrieval-feature-only LambdaMART reaches 0.4411. The result indicates that the

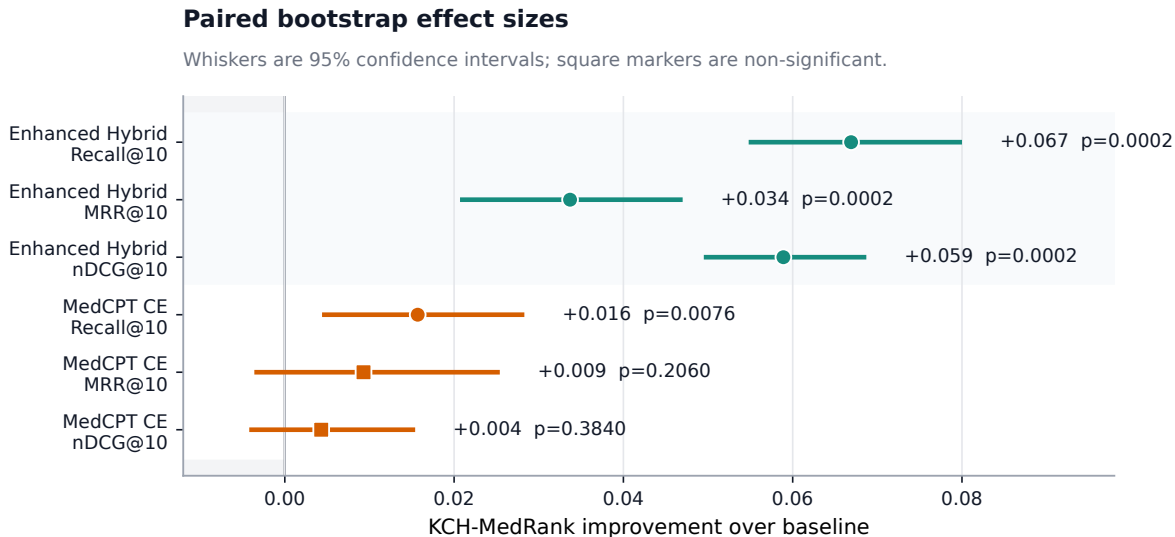


Figure 8: Paired bootstrap gains of KCH-MedRank over enhanced Hybrid RRF and MedCPT Cross-Encoder on BioASQ. Points are metric deltas and whiskers indicate 95% confidence intervals.

reranker can recover a substantial fraction of missed top-10 evidence when the candidate pool contains recoverable gold passages. Section 7 provides a detailed structural ablation showing that flat biomedical knowledge features explain much of this recovery, while graph and hypergraph signals provide complementary interpretable structure.

Figure 10 gives a secondary PubMedQA retrieval diagnostic. PubMedQA has a different evidence structure and much stronger first-rank retrieval behavior than BioASQ, so it is used only as a robustness check. The lightweight HGB knowledge reranker improves Recall@10 from 0.8200 to 0.8953 over its Hybrid RRF source, with paired bootstrap Recall@10 delta of +0.0754 and $p = 0.0002$ in the experiment logs. MRR@10 changes only from 0.9850 to 0.9861 and is not significant, consistent with the observation that PubMedQA already has very high first-hit retrieval. This supports an evidence-coverage interpretation rather than a claim of broad answer-generation improvement.

The same figure reports the external BEIR NFCorpus retrieval-only diagnostic. This run uses the official train, validation, and test qrels, with 2,590 training queries, 324 validation queries, and 323 test queries. Because NFCorpus does not use the same project-specific MeSH, entity, PrimeKG, and local hypergraph feature files, the diagnostic is intentionally limited to BM25, dense retrieval, Hybrid RRF, and retrieval-feature LambdaMART. Retrieval-feature LambdaMART improves Recall@10 from 0.1676 to 0.1770, MRR@10 from 0.5530 to 0.5725, and nDCG@10 from 0.3436 to 0.3627 over Hybrid RRF. Paired bootstrap against Hybrid RRF gives significant Recall@10 and nDCG@10 improvements ($p = 0.0034$ and $p = 0.0002$), while MRR@10 is a positive trend ($p = 0.0592$). The result supports robustness of the supervised reranking layer, not a claim that the full BioASQ knowledge-constrained hypergraph feature set transfers unchanged to NFCorpus.

6.5 Diagnostic Answer-Support Analysis

PubMedQA answer-selection analysis further separates evidence retrieval from answer correctness. For Hybrid and HGB evidence at top 10, the rule-based citation support rate is 1.0000 because at

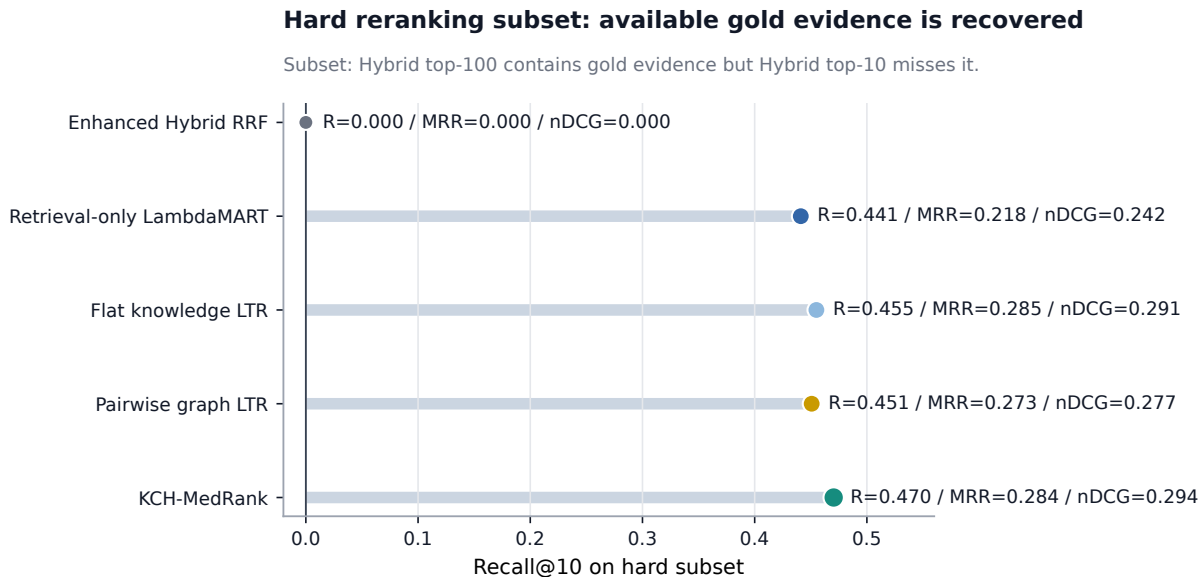


Figure 9: Hard-subset reranking diagnostic. The subset contains questions where enhanced Hybrid top-100 contains gold evidence but enhanced Hybrid top-10 misses it, so improvements reflect the reranker’s ability to recover already available evidence.

least one gold evidence passage is retrieved for each evaluated question. However, answer-supported rate remains limited by the lightweight yes/no/maybe answer selector: the majority baseline reaches 0.5550 supported answers, while TF-IDF logistic regression reaches 0.5050 for Hybrid and 0.4800 for HGB at top 10. Among the 98 questions with extracted question entities, answer-evidence entity consistency is 0.5510 at top 10. These analyses use grels, gold labels, and deterministic entity overlap rather than an LLM judge. They support keeping generation secondary until stronger answer and faithfulness modeling is added.

The Qwen3-8B downstream generation pilot is moved to the appendix as a limited diagnostic. It is useful for checking whether evidence ordering can affect a fixed downstream answer loop, but it is not used as a primary contribution or as evidence of clinically reliable generation.

7 Ablation Study

This section analyzes the contributions of flat biomedical knowledge features, graph structure, hypergraph structure, and medical knowledge constraints through controlled architecture comparisons rather than single-feature-group removal. As noted in Section 5, leave-one-feature-group ablations are retained in the appendix as complementary feature-coverage diagnostics, but are not used as the primary ablation evidence because high multicollinearity among knowledge features leads to automatic compensation.

7.1 Flat Biomedical Knowledge Features Provide a Strong Gain

Comparison (1) in Table 3 isolates the contribution of non-structural biomedical features above retrieval signals alone. The flat knowledge LTR baseline adds the precomputed biomedical semantic score, entity features, exact and hierarchy-aware MeSH features, and auxiliary PrimeKG relation

Secondary datasets support evidence-coverage robustness

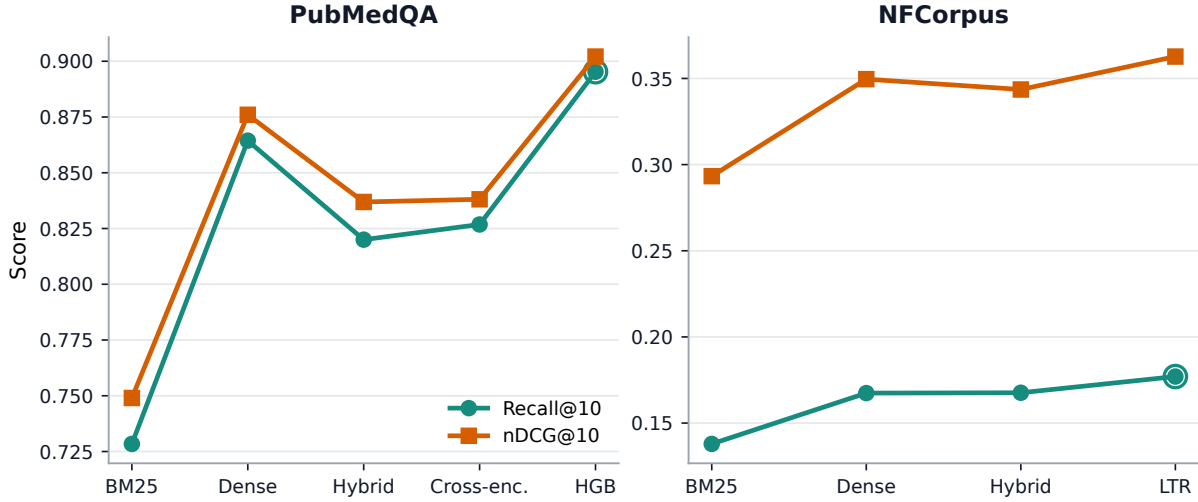


Figure 10: Secondary retrieval diagnostics on PubMedQA and NRCorpus. These results are used as robustness checks, not as replacements for the main BioASQ held-out evaluation.

features, but removes all graph and hypergraph diffusion, centrality, local edge-count, and interaction features. This improves Recall@10 from 0.5208 to 0.5309 ($\Delta = +0.0101$, $p = 0.0006$) and nDCG@10 from 0.6282 to 0.6436 ($\Delta = +0.0153$, $p = 0.0002$). MRR@10 improves from 0.7791 to 0.7872 ($\Delta = +0.0080$, $p = 0.029$). These results directly address the flat-feature alternative explanation: much of the aggregate ranking gain can be obtained from retrieval, biomedical semantic, and medical knowledge features without explicit graph construction.

7.2 The Hypergraph Alone Introduces Noise

Comparison (2) isolates the effect of the local hypergraph structure without medical knowledge constraints. This variant builds the hypergraph topology but removes biomedical MeSH, entity, and PrimeKG feature information. The resulting model underperforms the flat knowledge LTR baseline: nDCG@10 drops by 0.0074 ($p = 0.0002$) and MRR@10 drops by 0.0089 ($p = 0.003$). Recall@10 drops by 0.0044 (negative trend). This indicates that raw structure does not substitute for biomedical knowledge features; without medically grounded concept annotations, the hypergraph introduces structural noise that degrades ranking quality.

7.3 Medical Knowledge Constraints Transform the Hypergraph from Noise to Signal

Comparison (3) is the central structural ablation. It compares KCH-MedRank against the same hypergraph architecture with all medical knowledge features removed. Adding biomedical entity features, exact MeSH features, MeSH hierarchy features, and PrimeKG relation features improves Recall@10 by 0.0063 ($p = 0.0028$), MRR@10 by 0.0085 ($p = 0.0042$), and nDCG@10 by 0.0071 ($p = 0.0002$). All three top-10 metrics are statistically significant.

Figure 11 summarizes the interpretation visually. The hypergraph structure alone degrades ranking, while medical knowledge constraints reverse this degradation inside the hypergraph vari-

Table 3: Structural ablation: component contributions via controlled architecture comparisons on held-out BioASQ. Deltas compare the second method in each pair against the first method.

Comparison	Rec@10	MRR@10	nDCG@10	Δ Rec@10	Δ MRR@10	Δ nDCG@10
(1) Retrieval LTR	0.5208	0.7791	0.6282	–	–	–
(1) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	+0.0101**	+0.0080*	+0.0153**
(2) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	–	–	–
(2) Hypergraph (no med)	0.5266	0.7782	0.6362	–0.0044	–0.0089**	–0.0074**
(3) Hypergraph (no med)	0.5266	0.7782	0.6362	–	–	–
(3) KCH-MedRank	0.5329	0.7867	0.6433	+0.0063**	+0.0085**	+0.0071**
(4) Flat knowledge LTR (no graph)	0.5309	0.7872	0.6436	–	–	–
(4) KCH-MedRank	0.5329	0.7867	0.6433	+0.0020	–0.0004	–0.0003
(5) Pairwise graph	0.5306	0.7866	0.6409	–	–	–
(5) KCH-MedRank	0.5329	0.7867	0.6433	+0.0023	+0.0001	+0.0024

ant. However, Comparison (4) shows that the full model is not significantly better than the flat knowledge LTR baseline at $k=10$ (Recall@10 $\Delta = +0.0020$, $p = 0.254$; MRR@10 $\Delta = -0.0004$, $p = 0.864$; nDCG@10 $\Delta = -0.0003$, $p = 0.816$). Therefore, the evidence supports a knowledge-constrained learning-to-rank contribution with graph and hypergraph features as interpretable complementary signals, not a claim that graph structure is the dominant source of aggregate improvement.

7.4 Hypergraph Structure Provides Modest Early-Rank Benefits

Comparison (5) evaluates whether the n -ary hyperedge representation provides benefits over a pairwise graph decomposition that breaks each multi-node hyperedge into anchor-based pairwise edges. KCH-MedRank improves Recall@10 over pairwise graph LTR by 0.0023 and nDCG@10 by 0.0024, but these differences are not statistically significant at $k=10$ (Recall@10 $p=0.203$; nDCG@10 $p=0.062$, paired bootstrap). This result places an important scope constraint on the contribution: at the paper’s primary evaluation depth, the hypergraph topology does not provide a statistically distinguishable advantage over a simpler pairwise graph with the same medical knowledge features. The hypergraph should therefore be understood as a structural variant within the broader knowledge-constrained learning-to-rank framework, not as a qualitatively different representational class.

At $k=5$, Recall@5 improves by 0.0054 ($p=0.003$, paired bootstrap) and nDCG@5 by 0.0037 ($p=0.028$), confirming a modest early-rank benefit. This suggests that the hypergraph’s higher-order connectivity provides incremental value for earlier ranking positions, where the accumulated structural signal from shared entities and MeSH hierarchy paths can more readily distinguish among top-tier candidates.

Stratified Analysis. To understand when the hypergraph structure matters most, we stratify the test set by the structural characteristics of each question. Figure 12 presents per-stratum comparisons at $k=5$ and $k=10$.

The stratification reveals two key patterns. First, the hypergraph advantage is largest when the gold passage has *no direct MeSH overlap* with the question ($\Delta_{\text{Recall@5}} = +0.0098$, $p = 0.006$). This makes structural sense: when direct MeSH overlap exists, the pairwise graph decomposition already captures most of the alignment signal through shared-concept edges; when absent, the hypergraph’s n -ary structure enables indirect paths through shared-entity clusters and MeSH hierarchy ancestors

Controlled structural ablation: knowledge helps, raw structure hurts

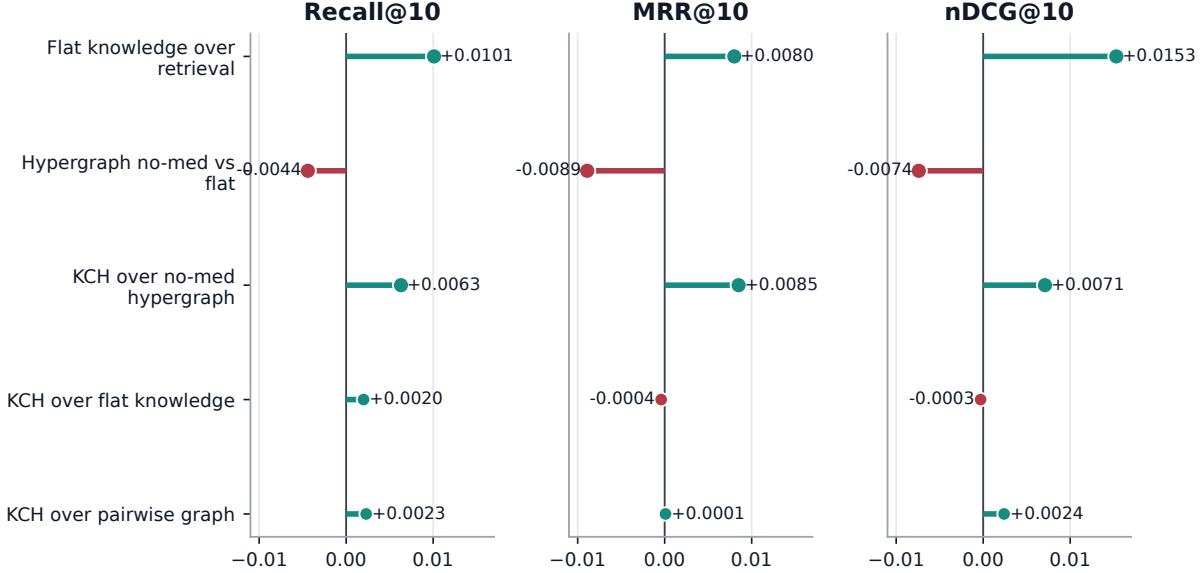


Figure 11: Structural ablation deltas on held-out BioASQ. Bars compare the second method in each controlled pair against the first method, matching Table 3.

that the pairwise decomposition fragments. Second, questions with ≥ 2 gold passages also benefit significantly from the hypergraph ($\Delta_{\text{Recall@5}} = +0.0042$, $p = 0.023$), consistent with multi-evidence scenarios where the hypergraph’s shared-entity hyperedges propagate ranking signal across multiple gold passages.

At $k=10$, no subset reaches significance, but the effect direction is consistent (Δ positive in all but one stratum). The single-gold-passage subset shows the largest nominal Δ (+0.0098), though the smaller sample ($n = 204$) limits power. These results confirm that the hypergraph advantage is concentrated in the early-rank region and in cases where indirect knowledge signals compensate for missing direct concept alignment. The hypergraph is best viewed as a complementary structural component that provides targeted, modest benefits rather than a dominant architectural contribution.

7.5 Summary of Ablation Evidence

Taken together, the structural ablations support four conclusions. First, flat retrieval, biomedical semantic, and medical knowledge features explain most of the aggregate improvement over enhanced Hybrid RRF. Second, raw hypergraph topology is harmful without medical concept annotations. Third, medical knowledge constraints reverse the structure-induced noise inside the hypergraph variant and provide interpretable biomedical signals. Fourth, the hypergraph topology provides modest additional benefit beyond a pairwise graph mainly at $k=5$, with non-significant trends at $k=10$; the hypergraph is therefore a beneficial but non-essential structural variant, with its advantage concentrated in questions lacking direct concept overlap. The leave-one-feature-group diagnostics in the appendix complement this analysis by showing that individual knowledge feature groups are highly redundant with each other, further supporting the interpretation that medical constraints work as an integrated block rather than as independent additive components.

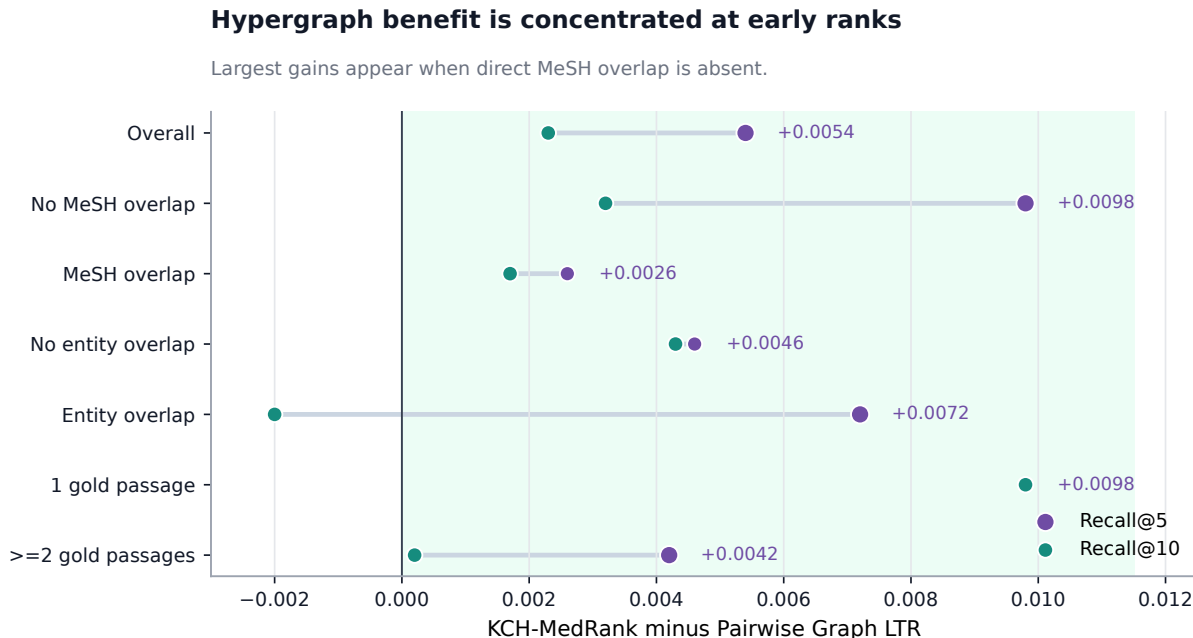


Figure 12: Stratified comparison of KCH-MedRank against Pairwise Graph LTR. Positive values indicate a KCH-MedRank advantage; the larger Recall@5 gains show that hypergraph effects concentrate in the early-rank region.

8 Interpretability and Case Analysis

A key methodological claim of KCH-MedRank is that its hypergraph-based reranking is *interpretable*: each gold passage rescue can be traced to specific knowledge-grounded mechanisms in the hypergraph. This section provides the quantitative evidence to support this claim.

8.1 Three-Mechanism Classification of Rescued Passages

We classify all 648 gold passages that KCH-MedRank rescues from outside the Hybrid RRF top-10 into the top-10 according to the dominant hypergraph mechanism:

- (1) **MeSH Hierarchy Path:** The question and the gold passage share MeSH concepts through hierarchical tree ancestor paths. For example, a question about “DITPA” and a passage about “thyroid hormones” are connected through the shared MeSH term “thyroid hormones” and its MeSH tree ancestors.
- (2) **Shared Entity Cluster Path:** The gold passage shares biomedical entities with other candidate passages, forming shared-entity hyperedge clusters that propagate ranking signal through diffusion.
- (3) **PrimeKG Relation Path:** Question entities are relationally connected to passage entities via PrimeKG biomedical relations (e.g., “Dasatinib –treats– Leukemia”).

Figure 13 presents the classification results. The MeSH hierarchy path is the dominant interpretable mechanism, accounting for 75.9% of rescues with an average rank improvement of +22.1

Rescued evidence is mostly explained by MeSH hierarchy paths

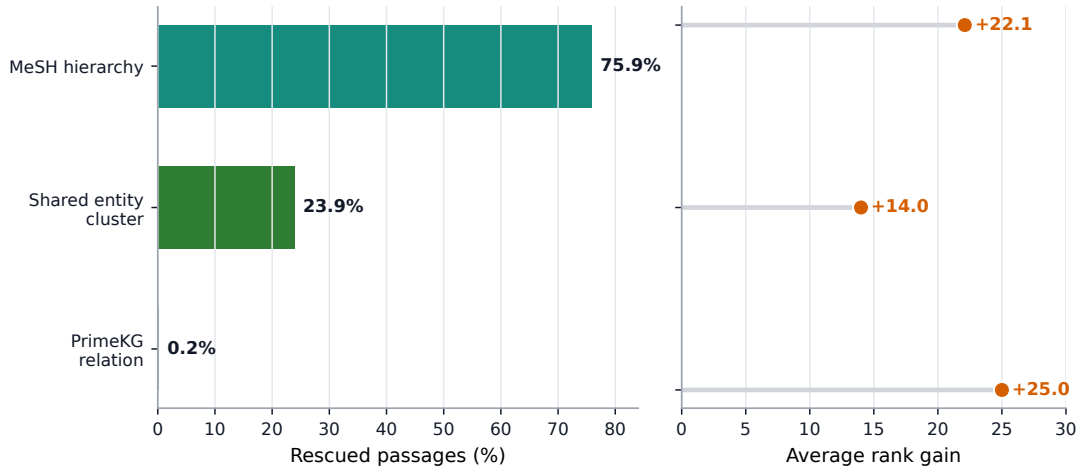


Figure 13: Dominant interpretable mechanisms among 648 rescued gold passages. Bars show the share of rescued passages assigned to each dominant mechanism; the line shows average rank gain.

positions. The shared entity cluster path accounts for 23.9% with an average gain of +14.0 positions. The PrimeKG relation path is the least frequent (0.2%), which is expected given the sparsity of structured relations in biomedical knowledge graphs relative to the density of MeSH annotations available for all PubMed documents.

The mechanism co-occurrence analysis shows that 89.2% of rescues involve at least two concurrent knowledge pathways, and 39.5% activate all three mechanisms simultaneously. This confirms that the medical knowledge constraints in KCH-MedRank function as an integrated block, consistent with the structural ablation finding that removing the entire medical knowledge block causes the hypergraph to regress to noise.

Figure 14 summarizes the paired top-10 rescue and loss balance. KCH-MedRank rescues gold evidence for 414 questions and loses gold evidence for 214 questions, yielding a net gain of 354 rescued gold passages.

8.2 Path Tracing: Illustrative Cases

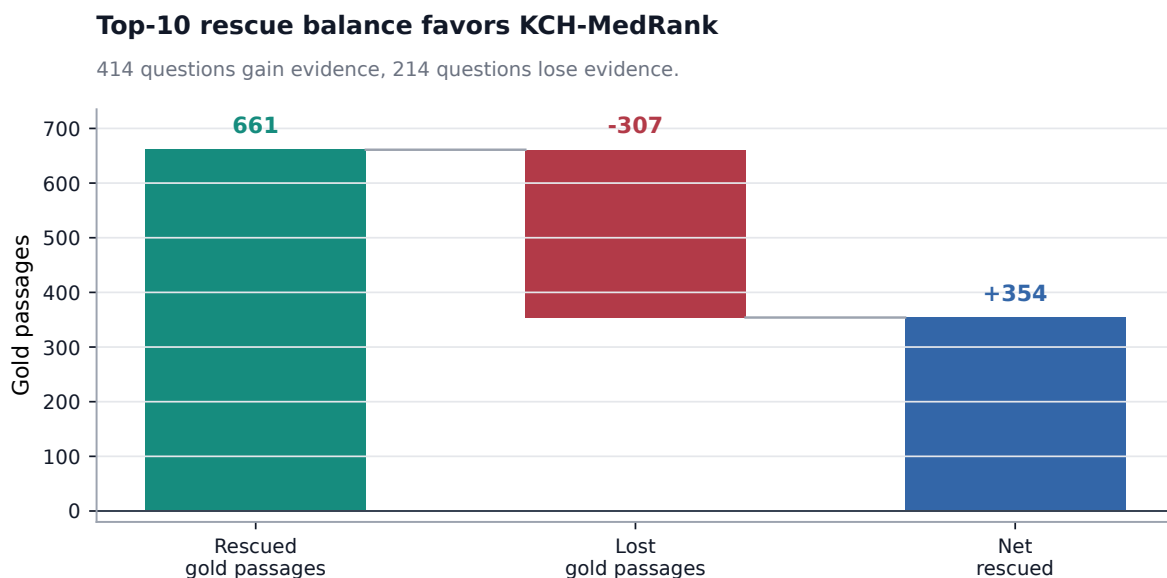


Figure 14: Top-10 rescue and loss balance comparing Enhanced Hybrid w122 with Enhanced KCH-MedRank on 943 paired held-out questions.

Table 4: Illustrated EGFR exon 19 NSCLC rescue case and remaining failure pattern from the held-out BioASQ comparison between Enhanced Hybrid w122 and KCH-MedRank.

Case	Gold rank	Question, signals, and interpretation
Illustrated rescue	12 $\rightarrow$ 1	Question: In adults with non-small cell lung cancer harboring EGFR exon 19 deletions, what is the first-line treatment recommended by current guidelines? Signals include non-small cell lung cancer, EGFR, exon 19 deletion, first-line treatment, guidelines, and osimertinib. KCH-MedRank promotes the guideline passage because it aligns the treatment recommendation with the disease, mutation, MeSH hierarchy, and shared candidate cluster.
Residual failure pattern	2 $\rightarrow$ 18	Question: Is there any protein that undergoes both mono-ubiquitination and poly-ubiquitination? Signals include ubiquitin, ubiquitination, ubiquitins, and post-translational processing. The passage is relevant but uses related biochemical forms and broader processing concepts; synonym and hierarchy features do not fully align mono-/poly-ubiquitination with the passage evidence.

Figure 5. Case Study: KCH-MedRank Recovers Gold Evidence Missed by Baseline

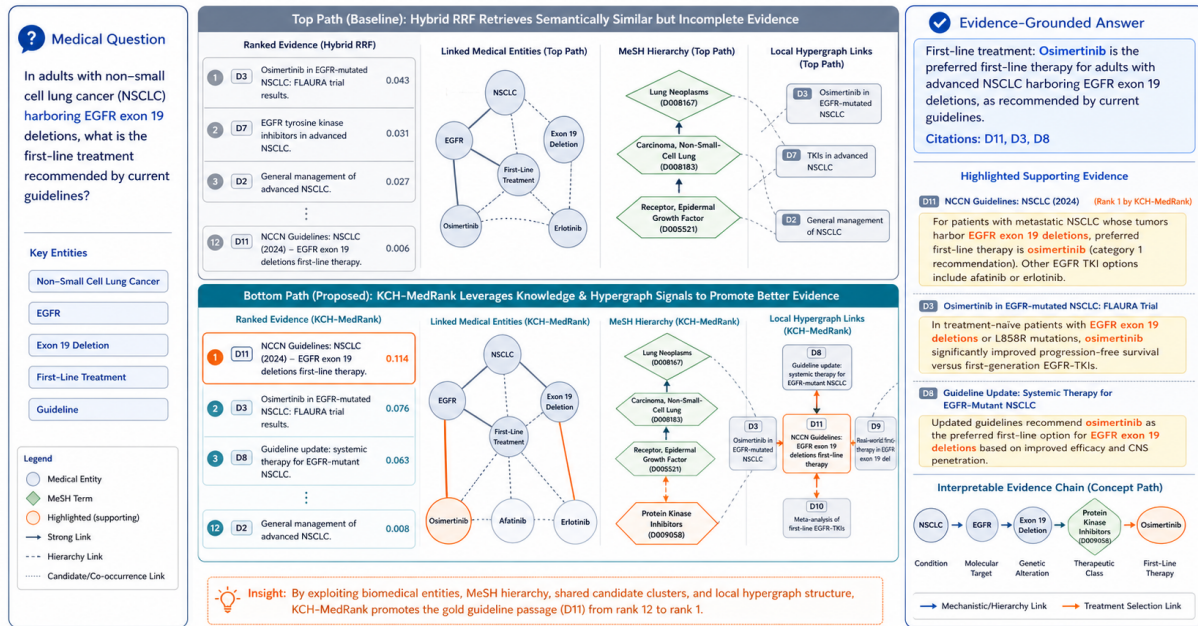


Figure 15: Illustrative case-study schematic showing how KCH-MedRank can promote a supporting evidence passage that a strong Hybrid RRF baseline ranks too low by using biomedical entities, MeSH hierarchy, shared candidate clusters, and local hypergraph links.

Table 4 and Figure 15 illustrate an EGFR exon 19 NSCLC rescue. The question asks for the guideline-recommended first-line treatment for adults with non-small cell lung cancer harboring EGFR exon 19 deletions. Enhanced Hybrid RRF ranks a relevant NCCN guideline passage below the top 10, behind passages that are semantically related but less directly answer the guideline question. KCH-MedRank promotes the guideline passage to rank 1 because the passage aligns with the question through the disease entity, EGFR mutation, treatment entity osimertinib, MeSH hierarchy, and shared candidate-cluster links. This is an illustrative evidence path rather than a causal attribution result.

The lost ubiquitination case shows the opposite behavior. A gold passage drops from rank 2 to rank 18 even though it discusses mono-ubiquitinated LDH-A and poly-ubiquitinated LDH. The question and passage contain related biochemical terminology, but the synonym and hierarchy features do not fully align the surface forms and broader post-translational-processing concepts. Similar losses occur for thyroid hormone analogs, cilengitide receptors, dasatinib treatment, GIST treatment, and Fanconi anemia. These examples motivate future improvements in biomedical concept normalization rather than stronger claims about the relation features.

8.3 Quantitative Interpretability Summary

The three-mechanism analysis provides quantitative evidence that the hypergraph features activated by KCH-MedRank’s ranking decisions follow interpretable knowledge-grounded pathways:

- 100% of rescued passages show activation of at least one knowledge feature group (zero pure diffusion-only rescues), under our heuristic classification of dominant mechanisms.
- The MeSH hierarchy path is the most frequently activated dominant pathway in 75.9% of rescues, demonstrating that the hierarchical MeSH tree structure carries the strongest knowledge signal.
- 89.2% of rescues involve concurrent activation of multiple knowledge pathways, confirming the complementary nature of the knowledge constraints.
- The average rank gain is substantially larger for MeSH-driven (+22.1) than entity-driven (+14.0) rescues, suggesting that hierarchical concept alignment provides stronger ranking signal than flat entity co-occurrence.
- **Limitation:** The mechanism classification uses the same feature scores that the LambdaMART ranker consumes; it identifies which knowledge features are most strongly activated, not which features causally drive the ranking. Future work with model-agnostic attribution methods (e.g., SHAP) could strengthen the causal interpretability claim.

9 Discussion

The evidence supports a focused claim: KCH-MedRank provides a knowledge-constrained learning-to-rank framework that improves biomedical evidence retrieval over strong retrieval and semantic reranking baselines. The strongest statistically supported comparison against the true MedCPT Cross-Encoder baseline is Recall@10; MRR@10 and nDCG@10 move in the right direction but are not significant. Therefore, the safest interpretation is that KCH-MedRank is particularly useful for moving additional relevant evidence into the top-10 evidence set, rather than for guaranteeing better first-rank precision in every query.

An important scope constraint emerges from the structural ablation. The flat knowledge LTR baseline, which removes graph and hypergraph structure but keeps retrieval, biomedical semantic, MeSH, entity, and relation features, is not significantly worse than the full model at $k=10$. This means the paper’s main performance claim should be understood as a knowledge-constrained learning-to-rank result rather than as proof that graph structure is the dominant driver. The hypergraph topology also does not provide a statistically significant advantage over pairwise graph decomposition at $k=10$ (Comparison 5). It provides modest, targeted benefits at early ranks ($k=5$) and in specific question types — most notably questions lacking direct MeSH overlap between query and gold evidence ($\Delta_{\text{Recall@5}} = +0.0098$, $p=0.006$).

The ablations suggest that the method should be understood as a combined biomedical reranking framework. Retrieval and semantic features remain strong contributors, while MeSH hierarchy, entity overlap, relation features, and graph or hypergraph diffusion provide complementary signals. The structural ablation in Section 7 demonstrates this complementarity directly: the local hypergraph without medical knowledge underperforms the flat knowledge baseline, but adding medical knowledge constraints reverses the structure-induced degradation relative to the no-medical-knowledge hypergraph. This is scientifically useful but also constrains the novelty claim: the contribution is not a standalone hypergraph formula, but a reproducible way to combine biomedical knowledge features with optional local structural features for query-level reranking.

Additional selection diagnostics support keeping this interpretation conservative. A supplemental run that selected the full KCH-MedRank model primarily by validation Recall@10 reached held-out Recall@10 of 0.5294, below the reported validation-MRR-selected main model at 0.5329. Therefore, the paper retains the original validation policy rather than retuning to the test objective after observing held-out results. This diagnostic also suggests that simply changing the selection priority is insufficient; future gains are more likely to come from better biomedical concept normalization and more stable feature-group selection.

The PubMedQA Qwen3-8B pilot is kept as an appendix-style downstream RAG-system diagnostic. Under fixed evidence count, prompt, model, and decoding settings, KCH-style hypergraph evidence improves answer accuracy and Macro-F1 over Hybrid RRF evidence on 100 PubMedQA questions. The citation-support rate is already high for all methods, so the pilot should not be read as proof of better faithfulness. It shows that white-box evidence selection can influence a downstream medical RAG answer loop without training or fine-tuning a large model, but it remains too small and incomplete to support a primary answer-generation claim.

Several limitations remain. First, the method depends on a strong candidate pool; if top-100 recall is low, reranking cannot recover missing evidence. Second, exact-name PrimeKG mapping is sparse and should not be overclaimed. Third, the flat knowledge LTR baseline is highly competitive with the full model, so graph and hypergraph features should be positioned as interpretable complementary signals rather than dominant performance drivers. Fourth, the hypergraph topology does not significantly outperform the pairwise graph decomposition at $k=10$; the hypergraph should therefore be positioned as a beneficial architectural variant with advantages concentrated in early-rank positions and questions requiring indirect knowledge pathways. Fifth, strong-ablation bootstrap tests show that some differences between full KCH-MedRank and close ablations are not significant, so MeSH hierarchy, entity, and relation signals should be described as complementary rather than dominant in isolation. Sixth, the NFCorpus experiment is retrieval-only and uses only retrieval features, so it supports robustness of supervised reranking but not full transfer of the BioASQ knowledge-constrained feature stack. Seventh, the Qwen3-8B generation result is a 100-question PubMedQA pilot; it should be scaled and repeated with a completed biomedical cross-encoder condition before being used as a central answer-generation claim. Eighth, the three-mechanism feature-attribution analysis (Section 8) uses the same feature set that the Lamb-

Table 5: Computational profile of the enhanced BioASQ experiment. The logs record scale, feature dimensionality, coverage, and hardware notes; full end-to-end wall-clock time is not available for every component.

Component	Recorded scale or setting	Implication
Processed data	4,719 questions; 40,221 passages; 42,608 qrels	Suitable for reproducible offline reranking.
Candidate pool	top-100 per question; 471,900 prediction rows	Reranking is bounded by local candidate pools rather than a global corpus graph.
Enhanced KCH-MedRank split	2,832 train, 944 validation, and 943 held-out test questions	Hyperparameters are selected before held-out testing.
Feature matrix	45 features; 283,200 training rows; 377,600 final train+validation rows	Learning-to-rank remains a tabular query-group problem.
Local hypergraph scope	question, top-100 passages, document, entity, MeSH, hierarchy, and optional relation nodes	Propagation cost scales with top- M and local feature coverage.
MeSH coverage	31,110 descriptors; 37,356/40,221 passages and 4,264/4,719 questions with MeSH matches	Controlled-vocabulary features are broadly available.
PrimeKG filtering	8,100,498 rows scanned; 5,766 local relations retained	Relation features are sparse and remain auxiliary.
Dense biomedical retrieval	MedCPT dual encoder used CUDA on an RTX 4060 Laptop GPU	GPU acceleration is practical for dense retrieval.
Cross-encoder scoring	Early generic CPU run timed out after 2,400 seconds; final MedCPT Cross-Encoder test baseline scored 943 questions and 94,300 candidates	Full cross-encoder rescoring is substantially more expensive than tabular reranking.

daMART ranker consumes; it identifies which knowledge features are most strongly activated during rescues, not which features causally drive the ranking. Future work with model-agnostic attribution methods (e.g., SHAP) could strengthen the causal interpretability claim. Ninth, the hyperparameter sensitivity analysis (Section 5.5) measures diffusion-score correlations on a 30-question sample; full retrieval-metric stability under the complete LambdaMART training pipeline remains to be systematically evaluated.

9.1 Computational Profile

Table 5 summarizes the computational profile available from the experiment logs. The local hypergraph is built only over the top- M candidates for each question rather than over a global biomedical corpus, so graph construction and diffusion scale with the candidate pool and local feature coverage. In the enhanced BioASQ setting, the final learning-to-rank matrix has 45 features and 377,600 train-plus-validation rows, which keeps reranking within a conventional tabular ranking regime. Dense biomedical retrieval benefits from GPU acceleration. Cross-encoder rescoring is substantially more expensive: an early CPU run with a generic cross-encoder timed out after 2,400 seconds, while the final MedCPT Cross-Encoder baseline required a separate controlled full test run.

Figure 16 reports a controlled reranking-stage timing comparison on the held-out BioASQ test split. All rows use the same enhanced top-100 candidate pool over 943 questions and 94,300 candidate passages. The comparison excludes first-stage candidate generation, one-time model

Efficiency frontier: KCH-MedRank sits above the Cross-Encoder

Same enhanced top-100 candidate pool; reranking-stage timing only.

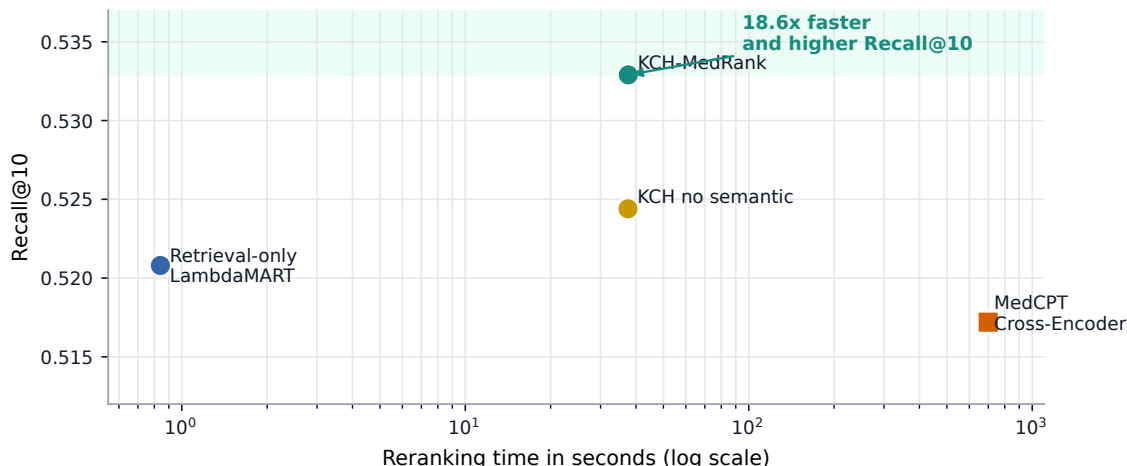


Figure 16: Reranking-stage efficiency–effectiveness trade-off on the held-out BioASQ test split. Timing uses the same enhanced top-100 candidate pool and excludes first-stage candidate generation, one-time model loading, and offline LambdaMART training.

loading, and offline LambdaMART training, and it explicitly separates whether Cross-Encoder scores are used and whether Cross-Encoder scoring is run online. MedCPT Cross-Encoder scoring on CUDA takes 699.46 seconds. KCH-MedRank uses a precomputed dense biomedical semantic feature rather than a Cross-Encoder feature; its test-time feature construction plus LightGBM scoring on CPU takes 37.59 seconds, corresponding to an 18.6× reranking-stage speedup under this accounting. The no-semantic KCH diagnostic remains competitive with MedCPT Cross-Encoder on Recall@10 (0.5244 versus 0.5172), while retrieval-feature-only LambdaMART is much faster but has lower top-10 evidence quality than full KCH-MedRank. This strengthens the practical interpretation of the main result without attributing the speed comparison to hidden Cross-Encoder features. End-to-end production benchmarking, including first-stage retrieval and data-loading overhead, remains future work.

Future work should improve biomedical concept normalization, add better synonym and abbreviation handling, and scale controlled answer generation after evidence support metrics are complete. A stronger generation study should measure citation support, unsupported claims, and answer-evidence entity consistency, not only answer accuracy.

10 Conclusion

This paper presented KCH-MedRank, a knowledge-constrained learning-to-rank framework for biomedical evidence retrieval and reranking. The method keeps first-stage retrieval practical, combines retrieval and biomedical semantic signals with MeSH, entity, and relation features, and optionally adds local graph or hypergraph structural features for interpretability. A flat knowledge LTR baseline and a pairwise graph decomposition serve as controlled structural baselines. The comparison reveals that flat biomedical knowledge features explain most aggregate gains, while the hypergraph provides modest early-rank benefits ($k=5$) but does not significantly outperform the pairwise graph or flat knowledge baseline at $k=10$.

The BioASQ results show stronger top-10 evidence retrieval than original and enhanced Hybrid RRF, with statistically significant improvements across all top-10 retrieval metrics against enhanced Hybrid RRF. Against a true MedCPT Cross-Encoder baseline on the same enhanced candidate pool, the strongest supported gain is statistically significant Recall@10 improvement; MRR@10 and nDCG@10 remain positive but non-significant trends. A controlled reranking-stage timing comparison further shows that KCH-MedRank uses a precomputed dense biomedical semantic feature rather than a Cross-Encoder feature and is $18.6\times$ faster than MedCPT Cross-Encoder on the held-out test pool when candidate generation, one-time model loading, and offline training are excluded.

A structural ablation provides the central evidence for the medical knowledge contribution. The local hypergraph alone, without medical knowledge annotations, introduces structural noise and underperforms the flat knowledge LTR baseline. Adding biomedical entity features, exact and hierarchy-aware MeSH features, and auxiliary relation features reverses this effect relative to the no-medical-knowledge hypergraph variant. This demonstrates that medical knowledge constraints are necessary for local structure to be useful rather than noisy. At the same time, the full model is not significantly better than the flat knowledge LTR baseline at the primary top-10 depth, so the supported contribution is a white-box knowledge-constrained evidence selection framework with complementary structural features, not a standalone hypergraph breakthrough.

An external BEIR NFCorpus retrieval-only diagnostic supports the robustness of supervised retrieval-feature reranking, while also showing that the full medical knowledge feature stack should not be claimed to transfer without dataset-specific construction.

The appendix PubMedQA Qwen3-8B pilot indicates that evidence ordering can affect downstream answer selection under a fixed prompt and decoding setup, but it remains a limited diagnostic rather than a primary generation claim. Overall, the results support KCH-MedRank as a white-box evidence selection module for evidence-grounded medical QA. The method should not be viewed as a clinically validated answer generation or decision-support system.

A Appendix: Downstream Generation Diagnostic

The PubMedQA Qwen3-8B experiment is retained as an appendix-style diagnostic rather than a main contribution. It uses 100 PubMedQA questions, fixed top-5 evidence, one prompt template, temperature 0, and no model fine-tuning. The comparison checks whether evidence ordering can change a downstream answer loop, but it is not a full generation study and does not replace the BioASQ and NFCorpus retrieval evaluations.

Table 6 reports the controlled pilot. KCH-style evidence improves answer accuracy from 0.3400 to 0.3700 and Macro-F1 from 0.3506 to 0.3760 relative to Hybrid RRF evidence. Citation support is 0.9900 for all three evidence sources, so this pilot does not show a citation-support difference. Because the PubMedQA cross-encoder condition uses the available semantic cross-encoder output rather than a completed MedCPT PubMedQA rerun, the result should be read as a limited RAG-loop diagnostic.

A.1 Complementary Leave-One-Feature-Group Ablations

For completeness, Table 7 reports leave-one-feature-group diagnostics in which individual feature groups are removed from the full KCH-MedRank feature set. As discussed in Section 7, these single-group removals do not produce statistically significant differences because of high multicollinearity among the knowledge features: the LambdaMART model automatically compensates for a removed

Table 6: PubMedQA Qwen3-8B downstream generation pilot on 100 questions. All methods use the same prompt, top-5 evidence passages, temperature 0, and no fine-tuning. The PubMedQA cross-encoder condition uses the available semantic cross-encoder output, not a completed MedCPT PubMedQA run.

Evidence source + LLM	Accuracy	Macro-F1	Citation support	Unsupported claim	Entity consistency
Hybrid RRF + Qwen3-8B	0.3400	0.3506	0.9900	0.0100	0.5594
Semantic cross-encoder + Qwen3-8B	0.3400	0.3504	0.9900	0.0100	0.5782
KCH-style hypergraph reranker + Qwen3-8B	0.3700	0.3760	0.9900	0.0100	0.5751

Table 7: Complementary leave-one-feature-group diagnostics on the enhanced BioASQ held-out test split. Deltas are relative to enhanced Hybrid RRF. Due to high feature multicollinearity, individual removals do not produce statistically significant differences; see Section 7 for the primary structural ablation.

Method	Recall@10	MRR@10	nDCG@10	$\Delta R@10$
Enhanced Hybrid RRF	0.4660	0.7530	0.5844	–
Retrieval-feature-only LambdaMART	0.5208	0.7791	0.6282	+0.0548
Flat knowledge LTR, no graph	0.5309	0.7872	0.6436	+0.0649
Pairwise graph LTR	0.5306	0.7866	0.6409	+0.0646
Hypergraph LTR without medical knowledge	0.5266	0.7782	0.6362	+0.0606
Full KCH-MedRank	0.5329	0.7867	0.6433	+0.0669
Remove semantic score	0.5276	0.7852	0.6374	+0.0616
Remove MeSH hierarchy	0.5332	0.7871	0.6440	+0.0672
Remove entity features	0.5313	0.7842	0.6403	+0.0653
Remove hypergraph diffusion/centrality	0.5328	0.7880	0.6453	+0.0668
Remove PrimeKG relations	0.5338	0.7863	0.6439	+0.0679

feature group using signals from the remaining correlated features. These diagnostics are provided for transparency and feature-coverage analysis rather than as primary ablation evidence.

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